EO-SAR Change Detection using Siamese U-Net

Technical Report

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ABSTRACT

Remote sensing-based change detection plays a crucial role in disaster assessment, infrastructure monitoring, environmental analysis, and urban planning. Traditional optical imagery-based approaches often suffer from illumination changes, cloud cover, and weather dependency. Synthetic Aperture Radar (SAR) imagery provides weather-independent sensing capabilities but introduces challenges due to speckle noise and modality differences.

This project presents a deep learning-based multimodal EO-SAR change detection framework using a Siamese U-Net architecture with a shared ResNet18 encoder. The proposed approach aims to detect binary structural changes between pre-event Electro-Optical (EO) imagery and post-event SAR imagery. The model extracts semantic features from both modalities using shared encoders and computes feature-level differences to identify regions of change.

A hybrid BCE + Dice Loss was employed to address severe class imbalance in the dataset. Additionally, valid-region masking was introduced during inference to suppress false positives in no-data triangular regions caused by sensor coverage inconsistencies.

The model achieved an IoU score of 0.2092 and an F1-score of 0.3460 on the provided test dataset. Experimental analysis demonstrated that the model successfully learned meaningful spatial change patterns despite challenges such as EO-SAR modality mismatch, sparse change regions, and SAR texture ambiguity.

TABLE OF CONTENTS

1. INTRODUCTION

Change detection in satellite imagery is an important problem in remote sensing and geospatial intelligence. It involves identifying changes between two images captured at different times. Applications include:

- Disaster damage assessment
- Urban expansion monitoring
- Infrastructure analysis
- Environmental monitoring
- Military surveillance
- Flood and wildfire assessment

Traditional change detection techniques rely heavily on optical imagery. However, optical sensors suffer from several limitations such as:

- cloud obstruction
- varying illumination
- weather dependency

Synthetic Aperture Radar (SAR) imagery provides an alternative because it:

- operates in all weather conditions
- works during both day and night
- penetrates clouds and smoke

However, EO and SAR imagery belong to different sensing modalities and exhibit highly different visual characteristics. This creates a challenging multimodal learning problem.

This project aims to solve EO-SAR binary change detection using deep learning-based semantic segmentation.

2. PROBLEM STATEMENT

The objective of this project is to perform binary change detection using:

- Pre-event EO satellite images
- Post-event SAR satellite images

The system predicts:

- A binary segmentation mask indicating changed and unchanged regions.

The major challenges include:

- EO-SAR modality mismatch
- Severe class imbalance
- Sparse change regions
- Invalid no-data areas
- SAR texture ambiguity

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3. OBJECTIVES

The main objectives of the project are:

- Develop an EO-SAR multimodal change detection pipeline
- Design a Siamese deep learning architecture
- Perform binary semantic segmentation
- Evaluate performance using IoU, Precision, Recall, and F1-score
- Analyze prediction failures and limitations
- Build a reproducible and modular inference pipeline

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4. LITERATURE SURVEY

4.1 Change Detection in Remote Sensing

Remote sensing change detection has traditionally been performed using:

- image differencing
- thresholding
- statistical methods
- feature matching

Recent advances in deep learning have enabled semantic segmentation-based change detection using Convolutional Neural Networks (CNNs).

4.2 Siamese Networks

Siamese architectures are commonly used for:

- image similarity
- feature comparison
- temporal change analysis

They use:

- shared encoders
- feature differencing
- semantic comparison

This makes them highly suitable for change detection tasks.

4.3 U-Net for Segmentation

U-Net is widely used in semantic segmentation because:

- encoder extracts semantic features
- decoder preserves spatial information
- skip connections improve localization

It is highly effective for pixel-level prediction tasks.

4.4 EO-SAR Multimodal Challenges

EO and SAR imagery differ significantly:

- EO captures optical texture and color
- SAR captures radar backscatter intensity

This creates:

- feature inconsistency
- domain gap
- texture ambiguity

Handling multimodal fusion is therefore a critical challenge.

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5. DATASET DESCRIPTION

The dataset contains:

- pre-event EO images
- post-event SAR images
- binary target masks

Dataset Structure

```
train/
├── pre-event/
├── post-event/
└── target/

val/
├── pre-event/
├── post-event/
└── target/

test/
├── pre-event/
├── post-event/
└── target/
```

Image Characteristics

Data Type	Shape
EO Image	1024 × 1024 × 3
SAR Image	1024 × 1024
Target Mask	1024 × 1024

Dataset Observations

Several important observations were made during data exploration:

1. Severe Class Imbalance

Approximately:

- 96% no-change pixels
- 4% change pixels

2. Invalid No-Data Regions

Many samples contained large black triangular regions caused by:

- sensor overlap mismatch
- geometric inconsistencies
- incomplete SAR coverage

3. Multimodal Differences

EO and SAR images showed highly different visual representations, increasing learning difficulty.

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6. METHODOLOGY

The proposed pipeline consists of:

1. Data preprocessing
2. EO-SAR feature extraction
3. Siamese feature comparison
4. Segmentation decoding
5. Binary mask prediction

6.1 Data Preprocessing

The following preprocessing steps were applied:

- TIFF image loading
- Image resizing to 256 × 256
- Normalization
- SAR channel expansion
- Tensor conversion

The SAR image was converted into pseudo 3-channel format for compatibility with pretrained CNN encoders.

6.2 Valid-Region Masking

During inference, invalid black regions were masked to suppress false positives.

This significantly improved:

- Precision
- IoU
- qualitative prediction quality

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7. MODEL ARCHITECTURE

The model uses a Siamese U-Net architecture with:

- shared ResNet18 encoder
- feature differencing
- U-Net decoder

Architecture Pipeline

EO Image
↓
Shared Encoder
↓
Feature Extraction

SAR Image
↓
Shared Encoder
↓
Feature Extraction

↓
Absolute Feature Difference
↓
U-Net Decoder
↓
Binary Change Mask

Why Siamese U-Net?

The architecture was chosen because:

- shared encoders learn consistent semantic representations
- feature differencing highlights changes
- U-Net preserves spatial localization
- lightweight and computationally efficient

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8. LOSS FUNCTIONS

A hybrid BCE + Dice Loss was used.

8.1 BCE Loss

Binary Cross Entropy handles:

- pixel-wise binary classification

8.2 Dice Loss

Dice Loss improves:

- overlap learning
- sparse region segmentation
- imbalance handling

Combined Loss

Total Loss = BCE Loss + Dice Loss

9. TRAINING CONFIGURATION

Parameter	Value
Image Size	256 × 256
Encoder	ResNet18
Optimizer	Adam
Learning Rate	0.0001
Batch Size	2
Epochs	2
Framework	PyTorch

10. EVALUATION METRICS

The following metrics were used:

IoU (Intersection over Union)

Measures overlap between prediction and ground truth.

Precision

Measures prediction correctness.

Recall

Measures ability to detect change regions.

F1 Score

Harmonic mean of precision and recall.

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11. EXPERIMENTAL RESULTS

Final Test Results

Metric	Score
IoU Score	0.2092
Precision	0.2907
Recall	0.4275
F1 Score	0.3460

Qualitative Results

The model successfully detected:

- structural boundaries
- major change regions
- spatial semantic differences

However, some false positives were observed in:

- invalid black regions
- high-texture SAR regions
- boundary areas

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12. CHALLENGES AND FAILURE ANALYSIS

12.1 EO-SAR Modality Gap

EO and SAR imagery exhibit highly different texture and intensity distributions.

This caused:

- feature mismatch
- semantic inconsistency
- prediction ambiguity

12.2 False Positives

The model frequently overpredicted changes in:

- invalid no-data regions
- bright SAR texture areas

This issue was mitigated using valid-region masking.

12.3 Sparse Change Regions

Small damaged regions were difficult to detect due to severe imbalance.

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13. FUTURE IMPROVEMENTS

Possible future enhancements include:

- Attention-based feature fusion
- Transformer-based encoders
- Multi-scale feature learning
- Advanced augmentation
- Better SAR denoising
- Domain adaptation techniques
- Self-supervised pretraining

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14. CONCLUSION

This project presented an EO-SAR multimodal change detection framework using Siamese U-Net architecture. Despite challenges such as modality mismatch, severe class imbalance, and invalid no-data regions, the model successfully learned meaningful change patterns.

The project demonstrated:

- multimodal feature learning
- semantic segmentation
- remote sensing AI pipeline development
- inference optimization
- failure analysis and debugging

The experimental results indicate that Siamese architectures combined with segmentation decoders provide a strong baseline for EO-SAR change detection tasks.

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